

Package #160

Embed an ANN-driven adaptive droop controller in the rooftop solar inverter that continuously modifies droop coefficients based on ambient temperature and load conditions to limit overcurrent and heat generation

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented problem 'thermal degradation' concerning rooftop solar inverter. User-submitted engineering problem (session input): "Why do commercial rooftop solar inverters fail prematurely in hot climates?". ROUTING: nhtsa_complaints: not queried — no vehicle (make|model|year) extracted from the problem; this source answers vehicle-parameterized questions only. [nhtsa_complaints] retrieval status NOT_QUERIED_GRAMMAR (no vehicle extracted; source grammar is make|model|year (routing decision, not absence)) — provider outcome recorded, not treated as absence (Art. XXI.3). [doe_osti] 10 records retrieved live (2026-2026). Illustrative records: 'Physics-of-Failure (PoF) based reliability assessment for photovoltaic (PV) inverters requires long-duration electrical and electrothermal stress histories, yet generating such stress histories with h'; 'Growth in power requirements and the complexity of emerging distribution systems are creating the need for new solid-state products for many applications. The qualification requirements, testing stand'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

Mechanism: Adaptive droop characteristic adjustment via ANN-assisted real-time tuning of droop coefficients and PI gains

Contents:

- 01 - Executive Technology Brief
- 02 - Engineering Technology-Transfer Dossier
- 03 - Buyer Decision Card
- 04 - Evidence Summary
- 05 - Transfer Manifest

Status: ENGINEERING_DEFINITION. Not physically validated.